

Control Systems (EEE/ECE/INSTR F242)

Tutorial 7 (24 February 2026)

1. The open loop transfer function of a system with unity feedback is given by $G(s) = K/(s(s+2))$. Determine the gain 'K' such that closed loop system has a damping ratio of 0.316. For this value of K, find rise time (0% to 100%), percentage overshoot, peak value of output, peak time and settling time (2% tolerance band), when the system is subjected to a step input of 12 units.
2. A unity-feedback closed loop control system is to be designed to meet the following specifications,
 $30\% > M_p(\%) > 15\%$
 t_s (2% tolerance band) < 0.75 sec
Mark the area in the s-plane in which the roots of the system's characteristic equation must lie. Assume the closed loop system is of second order.
3. For the system shown in Fig. Q3, determine the values of K and K_h , so that the maximum overshoot in the unit step response is 0.2 and the peak time is 1 sec. With these values of K and K_h , obtain the rise time and settling time. (assume $J=1$ kg-m² and $B=1$ N-m/rad/sec)

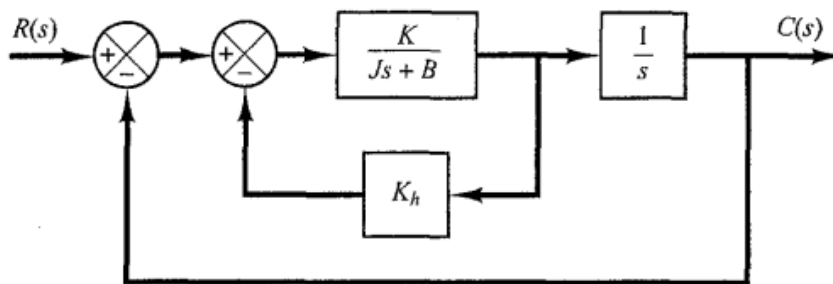


Fig. Q3
